

Statistical Machine Learning EndSem 2026

Rubrics

3rd May 2026

Q1. Random Forest

Question. Which of the following statements about Random Forest are correct?

- (A) It trains multiple decision trees using bootstrap samples of the training data.
- (B) It usually combines predictions from different trees using majority voting for classification.
- (C) It always trains only one decision tree.
- (D) It can reduce variance compared to a single decision tree.
- (E) It selects a random subset of features while splitting nodes.

Solution

A Random Forest is an ensemble of decision trees. Each tree is usually trained on a bootstrap sample of the data. For classification, predictions are commonly combined using majority voting. Random feature selection at split nodes decorrelates trees, and averaging across trees reduces variance compared to a single high-variance decision tree.

Therefore, the correct options are

A, B, D, E.

Rubric: 2.5 marks

Item	Marks
Correctly identifies A	0.5
Correctly identifies B	0.5
Correctly identifies D	0.5
Correctly identifies E	0.5

Q2. Principal Component Analysis

Question. Which of the following statements about PCA are correct?

- (A) PCA finds directions along which the projected data has maximum variance.
- (B) The first principal component corresponds to the direction of maximum variance in the data.
- (C) PCA always improves classification accuracy.
- (D) Principal components are mutually orthogonal.
- (E) PCA tries to maximize between-class separation.

Solution

PCA is an unsupervised dimensionality reduction method. It finds orthogonal directions of maximum variance. The first principal component is the direction along which the projected data has the largest variance. PCA does not use class labels and therefore does not explicitly maximize between-class separation. Also, PCA does not always improve classification accuracy.

Therefore, the correct options are

A, B, D.

Rubric: 2.5 marks

Item	Marks
Correctly identifies A	0.5
Correctly identifies B	0.5
Correctly identifies D	0.5

Q3. Fisher Discriminant Analysis

Question. Which of the following statements about FDA are correct?

- (A) FDA uses class labels to find a projection direction.
- (B) FDA ignores class labels and only maximizes total variance.
- (C) In the two-class case, FDA projects data onto a single discriminant direction.
- (D) FDA and PCA always produce the same projection direction.
- (E) FDA tries to make samples from the same class compact after projection.

Solution

FDA is a supervised method and uses class labels. In the binary case, classical FDA finds one discriminant direction. The Fisher criterion tries to make different class means well-separated after projection while keeping within-class scatter small. Hence it tries to make samples from the same class compact after projection.

Therefore, the correct options are

A, C, E.

Rubric: 2.5 marks

Item	Marks
Correctly identifies A	0.5
Correctly identifies C	0.5
Correctly identifies E	0.5

Q4. Discriminant Analysis

Question. Which of the following statements about Discriminant Analysis are correct?

- (A) Quadratic Discriminant Analysis can produce nonlinear decision boundaries.
- (B) Discriminant Analysis does not use class labels during training.
- (C) LDA commonly assumes that different classes share the same covariance matrix.
- (D) QDA allows different classes to have different covariance matrices.

Solution

LDA and QDA are supervised methods and use class labels during training. LDA assumes class-conditional Gaussian distributions with a common covariance matrix. QDA allows each class to have its own covariance matrix. Because of this, QDA can produce quadratic, and hence nonlinear, decision boundaries.

Therefore, the correct options are

A, C, D.

Rubric: 2.5 marks

Item	Marks
Correctly identifies A	0.5
Correctly identifies C	0.5
Correctly identifies D	0.5

Q5.

Question. Consider a binary classification dataset

$$D = \{(x_i, y_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^d, \quad y_i \in \{-1, +1\}.$$

We want to learn two linear hyperplanes. The normal vectors of both hyperplanes are assumed to be unit vectors. Write an objective function satisfying:

1. signed distance $d_1(x_i)$ is close to y_i ,
2. signed distance $d_2(x_i)$ is close to y_i ,
3. the two hyperplanes are orthogonal,
4. the unit-vector constraint need not be taken care of in the objective.

Write the objective function and update rule for all parameters of one of the hyperplanes.

Solution

Let the two hyperplanes be

$$d_1(x) = w_1^\top x + b_1,$$

and

$$d_2(x) = w_2^\top x + b_2.$$

Since w_1 and w_2 are assumed to be unit vectors, $d_1(x)$ and $d_2(x)$ may be interpreted as signed distances. We do not explicitly enforce

$$\|w_1\|_2 = 1, \quad \|w_2\|_2 = 1$$

in the objective because the question says that this constraint need not be handled.

The first requirement can be captured by

$$w_1^\top x_i + b_1 \approx y_i.$$

The second requirement can be captured by

$$w_2^\top x_i + b_2 \approx y_i.$$

Orthogonality of hyperplanes means orthogonality of normal vectors:

$$w_1^\top w_2 = 0.$$

A suitable objective is

$$J(w_1, b_1, w_2, b_2) = \frac{1}{2} \sum_{i=1}^n \left[(w_1^\top x_i + b_1 - y_i)^2 + (w_2^\top x_i + b_2 - y_i)^2 \right] + \frac{\lambda}{2} (w_1^\top w_2)^2$$

where $\lambda > 0$ controls the orthogonality penalty.

$w_1^\top w_2$ is also fine instead of squaring. In that case gradient wrt w_1 will be just w_2 .

Define

$$e_{1i} = w_1^\top x_i + b_1 - y_i.$$

The part of the objective involving w_1, b_1 is

$$J_1 = \frac{1}{2} \sum_{i=1}^n e_{1i}^2 + \frac{\lambda}{2} (w_1^\top w_2)^2.$$

The gradient with respect to w_1 is

$$\nabla_{w_1} J = \sum_{i=1}^n e_{1i} x_i + \lambda (w_1^\top w_2) w_2.$$

The gradient with respect to b_1 is

$$\frac{\partial J}{\partial b_1} = \sum_{i=1}^n e_{1i}.$$

Therefore, the gradient descent updates for the first hyperplane are

$$w_1 \leftarrow w_1 - \eta \left[\sum_{i=1}^n (w_1^\top x_i + b_1 - y_i) x_i + \lambda (w_1^\top w_2) w_2 \right]$$

and

$$b_1 \leftarrow b_1 - \eta \sum_{i=1}^n (w_1^\top x_i + b_1 - y_i).$$

Similarly, for the second hyperplane,

$$w_2 \leftarrow w_2 - \eta \left[\sum_{i=1}^n (w_2^\top x_i + b_2 - y_i) x_i + \lambda (w_1^\top w_2) w_1 \right],$$

and

$$b_2 \leftarrow b_2 - \eta \sum_{i=1}^n (w_2^\top x_i + b_2 - y_i).$$

Rubric: 4 marks

Item	Marks
Defines two signed hyperplane outputs $d_1(x) = w_1^\top x + b_1$, $d_2(x) = w_2^\top x + b_2$	0.5
Includes squared closeness terms to y_i for both hyperplanes	1.0
Correctly encodes orthogonality using $(w_1^\top w_2)^2$ or equivalent	1.0
Correctly derives gradient with respect to w_1 including orthogonality penalty	1.0
Correctly derives update for b_1	0.5

Q6.

Question. Consider the scalar regression dataset

$$(x_1, y_1) = (1, 1), \quad (x_2, y_2) = (2, 3), \quad (x_3, y_3) = (3, 5).$$

In gradient boosting, use

$$L(y, F(x)) = \exp(|y - F(x)|).$$

The initial prediction is the average of the target values.

Initial Model

The initial prediction is

$$F_0(x) = \frac{1 + 3 + 5}{3} = 3.$$

Thus,

$$F_0(x_1) = F_0(x_2) = F_0(x_3) = 3.$$

Q6(a): Negative Gradients

Let

$$r_i = y_i - F(x_i).$$

Then

$$L_i = \exp(|r_i|).$$

We need

$$-\frac{\partial L_i}{\partial F(x_i)}.$$

Since

$$r_i = y_i - F(x_i),$$

we have

$$\frac{\partial r_i}{\partial F(x_i)} = -1.$$

For $r_i \neq 0$,

$$\frac{\partial |r_i|}{\partial F(x_i)} = -\text{sign}(r_i).$$

Therefore,

$$\frac{\partial L_i}{\partial F(x_i)} = -\exp(|r_i|) \text{sign}(r_i).$$

Hence the negative gradient is

$$\boxed{-\frac{\partial L_i}{\partial F(x_i)} = \exp(|y_i - F(x_i)|) \text{sign}(y_i - F(x_i)).}$$

At $y_i = F(x_i)$, we use $\text{sign}(0) = 0$.

Q6(b): Modified Targets

At $F_0(x) = 3$, the residuals are

$$\begin{aligned}r_1 &= 1 - 3 = -2, \\r_2 &= 3 - 3 = 0, \\r_3 &= 5 - 3 = 2.\end{aligned}$$

The modified target is the negative gradient:

$$\tilde{y}_i = \exp(|r_i|) \operatorname{sign}(r_i).$$

Thus,

$$\tilde{y}_1 = -e^2, \quad \tilde{y}_2 = 0, \quad \tilde{y}_3 = e^2.$$

Therefore,

$$\boxed{(\tilde{y}_1, \tilde{y}_2, \tilde{y}_3) = (-e^2, 0, e^2)}.$$

Q6(c): Fit a One-Split Regression Tree

Let

$$E = e^2.$$

The training pairs for fitting the next tree are

$$(1, -E), \quad (2, 0), \quad (3, E).$$

We compare the two allowed splits.

Split 1: $x \leq 1.5$

Left region:

$$R_L = \{x = 1\},$$

with mean prediction

$$c_L = -E.$$

Right region:

$$R_R = \{x = 2, x = 3\},$$

with mean prediction

$$c_R = \frac{0 + E}{2} = \frac{E}{2}.$$

The squared error is

$$\text{SSE}_1 = (-E + E)^2 + \left(0 - \frac{E}{2}\right)^2 + \left(E - \frac{E}{2}\right)^2.$$

Thus,

$$\text{SSE}_1 = 0 + \frac{E^2}{4} + \frac{E^2}{4} = \frac{E^2}{2}.$$

Since $E = e^2$,

$$\boxed{\text{SSE}_1 = \frac{e^4}{2}}.$$

Split 2: $x \leq 2.5$

Left region:

$$R_L = \{x = 1, x = 2\},$$

with mean prediction

$$c_L = \frac{-E + 0}{2} = -\frac{E}{2}.$$

Right region:

$$R_R = \{x = 3\},$$

with mean prediction

$$c_R = E.$$

The squared error is

$$\text{SSE}_2 = \left(-E + \frac{E}{2}\right)^2 + \left(0 + \frac{E}{2}\right)^2 + (E - E)^2.$$

Thus,

$$\text{SSE}_2 = \frac{E^2}{4} + \frac{E^2}{4} + 0 = \frac{E^2}{2}.$$

Hence,

$$\boxed{\text{SSE}_2 = \frac{e^4}{2}}.$$

Both splits are equally good. Since the question says to use the smallest split value in case of a tie, we choose

$$\boxed{x \leq 1.5}.$$

Therefore,

$$\boxed{h_1(x) = \begin{cases} -e^2, & x \leq 1.5, \\ \frac{e^2}{2}, & x > 1.5. \end{cases}}$$

Q6(d): Updated Model

The gradient boosting update is

$$F_1(x) = F_0(x) + \rho h_1(x).$$

Given $\rho = 1$,

$$F_1(x) = 3 + h_1(x).$$

Therefore,

$$\boxed{F_1(x) = \begin{cases} 3 - e^2, & x \leq 1.5, \\ 3 + \frac{e^2}{2}, & x > 1.5. \end{cases}}$$

For $x_1 = 1$, since $1 \leq 1.5$,

$$F_1(1) = 3 - e^2.$$

Thus,

$$F_1(1) = 3 - e^2.$$

Rubric: 4 marks

Item	Marks
Derives correct negative gradient expression	1.0
Computes modified targets $(-e^2, 0, e^2)$. In case there is a minor calculation mistake, deduct 0.25	1.0
Computes SSEs for both splits and identifies tie. In case there is a minor calculation mistake, deduct 0.25	0.5
Uses tie-breaking rule to choose $x \leq 1.5$ and gives correct $h_1(x)$	0.5
Writes updated model $F_1(x) = 3 + h_1(x)$	0.5
Computes $F_1(1) = 3 - e^2$	0.5

Q7.

Consider a neural network with two input nodes x_1, x_2 , one hidden node, one output node, no bias terms, absolute-value activation at the hidden node, and no activation at the output node. The input-to-hidden weights are w_1, w_2 , and the hidden-to-output weight is w_3 . The loss is MSE. Find an update rule for w_1 .

Solution

For one training example (x_1, x_2, y) , the hidden pre-activation is

$$a = w_1x_1 + w_2x_2.$$

The hidden activation is

$$h = |a|.$$

The output is

$$\hat{y} = w_3h = w_3|w_1x_1 + w_2x_2|.$$

Using squared error loss,

$$L = (y - \hat{y})^2.$$

By the chain rule,

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial h} \frac{\partial h}{\partial a} \frac{\partial a}{\partial w_1}.$$

Now,

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y),$$

$$\frac{\partial \hat{y}}{\partial h} = w_3,$$

$$\frac{\partial h}{\partial a} = \text{sign}(a),$$

for $a \neq 0$, and

$$\frac{\partial a}{\partial w_1} = x_1.$$

Therefore,

$$\boxed{\frac{\partial L}{\partial w_1} = 2(\hat{y} - y)w_3 \text{sign}(a)x_1.}$$

Substituting $a = w_1x_1 + w_2x_2$, we get

$$\boxed{\frac{\partial L}{\partial w_1} = 2[w_3|w_1x_1 + w_2x_2| - y]w_3 \text{sign}(w_1x_1 + w_2x_2)x_1.}$$

Hence the gradient descent update is

$$\boxed{w_1 \leftarrow w_1 - \eta 2[w_3|w_1x_1 + w_2x_2| - y]w_3 \text{sign}(w_1x_1 + w_2x_2)x_1.}$$

If the MSE is defined as

$$L = \frac{1}{2}(y - \hat{y})^2,$$

then the update becomes

$$w_1 \leftarrow w_1 - \eta [w_3 |w_1 x_1 + w_2 x_2| - y] w_3 \text{sign}(w_1 x_1 + w_2 x_2) x_1.$$

At $a = 0$, use a subgradient:

$$\text{sign}(0) \in [-1, 1].$$

A common choice is $\text{sign}(0) = 0$.

Rubric: 4 marks

Item	Marks
Correctly defines hidden pre-activation $a = w_1 x_1 + w_2 x_2$ and hidden activation $h = a $	0.75
Correctly writes output $\hat{y} = w_3 a $	0.5
Correctly applies chain rule	1.0
Correctly handles derivative of absolute value using $\text{sign}(a)$ or subgradient	0.75
Writes final update rule for w_1	1.0

Q8.

Question. Show that the MLE estimate of variance of a Gaussian pdf is biased.

Solution

ML estimate of mean is unbiased.

We want to check whether the ML estimate of variance is unbiased, i.e.,

$$\mathbb{E} [\hat{\sigma}_{\text{ML}}^2] \stackrel{?}{=} \sigma^2.$$

The maximum likelihood estimate of variance is

$$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu}_{\text{ML}})^2,$$

where

$$\hat{\mu}_{\text{ML}} = \frac{1}{N} \sum_{i=1}^N x_i.$$

Now,

$$\mathbb{E} [\hat{\sigma}_{\text{ML}}^2] = \mathbb{E} \left[\frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu}_{\text{ML}})^2 \right].$$

Expanding the square,

$$\mathbb{E} [\hat{\sigma}_{\text{ML}}^2] = \frac{1}{N} \mathbb{E} \left[\sum_{i=1}^N (x_i^2 - 2x_i \hat{\mu}_{\text{ML}} + \hat{\mu}_{\text{ML}}^2) \right].$$

Therefore,

$$\mathbb{E} [\hat{\sigma}_{\text{ML}}^2] = \frac{1}{N} \mathbb{E} \left[\sum_{i=1}^N x_i^2 - 2\hat{\mu}_{\text{ML}} \sum_{i=1}^N x_i + \sum_{i=1}^N \hat{\mu}_{\text{ML}}^2 \right].$$

Since

$$\sum_{i=1}^N x_i = N\hat{\mu}_{\text{ML}},$$

we get

$$\mathbb{E} [\hat{\sigma}_{\text{ML}}^2] = \frac{1}{N} \mathbb{E} \left[\sum_{i=1}^N x_i^2 - 2N\hat{\mu}_{\text{ML}}^2 + N\hat{\mu}_{\text{ML}}^2 \right].$$

Thus,

$$\mathbb{E} [\hat{\sigma}_{\text{ML}}^2] = \frac{1}{N} \mathbb{E} \left[\sum_{i=1}^N x_i^2 - N\hat{\mu}_{\text{ML}}^2 \right].$$

So,

$$\mathbb{E} [\hat{\sigma}_{\text{ML}}^2] = \frac{1}{N} \sum_{i=1}^N \mathbb{E}[x_i^2] - \mathbb{E} [\hat{\mu}_{\text{ML}}^2].$$

Now,

$$\mathbb{E}[x_i^2] = \text{Var}(x_i) + (\mathbb{E}[x_i])^2.$$

Since

$$\text{Var}(x_i) = \sigma^2, \quad \mathbb{E}[x_i] = \mu,$$

we have

$$\mathbb{E}[x_i^2] = \sigma^2 + \mu^2.$$

Also,

$$\mathbb{E}[\hat{\mu}_{\text{ML}}^2] = \text{Var}(\hat{\mu}_{\text{ML}}) + (\mathbb{E}[\hat{\mu}_{\text{ML}}])^2.$$

Since

$$\hat{\mu}_{\text{ML}} = \frac{1}{N} \sum_{i=1}^N x_i,$$

we have

$$\mathbb{E}[\hat{\mu}_{\text{ML}}] = \mu.$$

Also,

$$\text{Var}(\hat{\mu}_{\text{ML}}) = \text{Var}\left(\frac{1}{N} \sum_{i=1}^N x_i\right).$$

Using independence,

$$\text{Var}\left(\frac{1}{N} \sum_{i=1}^N x_i\right) = \frac{1}{N^2} \text{Var}\left(\sum_{i=1}^N x_i\right) = \frac{1}{N^2} \sum_{i=1}^N \text{Var}(x_i).$$

Therefore,

$$\text{Var}(\hat{\mu}_{\text{ML}}) = \frac{1}{N^2} \cdot N\sigma^2 = \frac{\sigma^2}{N}.$$

Hence,

$$\mathbb{E}[\hat{\mu}_{\text{ML}}^2] = \frac{\sigma^2}{N} + \mu^2.$$

Substituting back,

$$\mathbb{E}[\hat{\sigma}_{\text{ML}}^2] = \frac{1}{N} \sum_{i=1}^N (\sigma^2 + \mu^2) - \left(\frac{\sigma^2}{N} + \mu^2\right).$$

Thus,

$$\mathbb{E}[\hat{\sigma}_{\text{ML}}^2] = \sigma^2 + \mu^2 - \frac{\sigma^2}{N} - \mu^2.$$

Therefore,

$$\mathbb{E}[\hat{\sigma}_{\text{ML}}^2] = \sigma^2 - \frac{\sigma^2}{N}.$$

Hence,

$$\boxed{\mathbb{E}[\hat{\sigma}_{\text{ML}}^2] = \frac{N-1}{N} \sigma^2}$$

Since

$$\frac{N-1}{N} \sigma^2 \neq \sigma^2,$$

we conclude that

$$\hat{\sigma}_{\text{ML}}^2 \text{ is a biased estimator of } \sigma^2.$$

The bias is

$$\text{Bias}(\hat{\sigma}_{\text{ML}}^2) = \mathbb{E}[\hat{\sigma}_{\text{ML}}^2] - \sigma^2.$$

Therefore,

$$\text{Bias}(\hat{\sigma}_{\text{ML}}^2) = \frac{N-1}{N}\sigma^2 - \sigma^2 = -\frac{\sigma^2}{N}.$$

So,

$$\text{Bias}(\hat{\sigma}_{\text{ML}}^2) = -\frac{\sigma^2}{N}$$

The unbiased estimator of variance is therefore

$$\hat{\sigma}_{\text{unbiased}}^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \hat{\mu}_{\text{ML}})^2.$$

Rubric: 4 marks

Item	Marks
Correctly writes the ML variance estimator	1
$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu}_{\text{ML}})^2$	
where	
$\hat{\mu}_{\text{ML}} = \frac{1}{N} \sum_{i=1}^N x_i$	
Correctly expands the squared term:	1.00
$(x_i - \hat{\mu}_{\text{ML}})^2 = x_i^2 - 2x_i\hat{\mu}_{\text{ML}} + \hat{\mu}_{\text{ML}}^2$	
and simplifies using	
$\sum_{i=1}^N x_i = N\hat{\mu}_{\text{ML}}$	
to obtain	
$\mathbb{E}[\hat{\sigma}_{\text{ML}}^2] = \frac{1}{N} \sum_{i=1}^N \mathbb{E}[x_i^2] - \mathbb{E}[\hat{\mu}_{\text{ML}}^2]$	
Correctly evaluates	1.00
$\mathbb{E}[\hat{\mu}_{\text{ML}}^2] = \text{Var}(\hat{\mu}_{\text{ML}}) + (\mathbb{E}[\hat{\mu}_{\text{ML}}])^2$	
with	
$\mathbb{E}[\hat{\mu}_{\text{ML}}] = \mu$	
and	
$\text{Var}(\hat{\mu}_{\text{ML}}) = \text{Var}\left(\frac{1}{N} \sum_{i=1}^N x_i\right) = \frac{\sigma^2}{N}$	
Correctly substitutes the above quantities to obtain	1
$\mathbb{E}[\hat{\sigma}_{\text{ML}}^2] = \sigma^2 + \mu^2 - \left(\frac{\sigma^2}{N} + \mu^2\right) = \frac{N-1}{N}\sigma^2$	

Q9.

In PCA, the first principal component direction is obtained by maximizing the variance of the projected data. Suppose we modify the PCA objective. Instead of maximizing the sum of squared projections, we maximize the sum of absolute projections while maintaining the constraint that the projection vector $u \in \mathbb{R}^d$ is unit vector. Find u .

Solution Using Lagrangian

The modified objective is

$$\max_{u \in \mathbb{R}^d} \sum_{i=1}^n |u^\top x_i|$$

subject to

$$u^\top u = 1.$$

Define

$$J(u) = \sum_{i=1}^n |u^\top x_i|.$$

The Lagrangian is

$$\mathcal{L}(u, \lambda) = \sum_{i=1}^n |u^\top x_i| - \lambda(u^\top u - 1).$$

$$\frac{\partial}{\partial u} |u^\top x_i| = \text{sign}(u^\top x_i) x_i.$$

Therefore,

$$\nabla_u \mathcal{L} = \sum_{i=1}^n \text{sign}(u^\top x_i) x_i - 2\lambda u.$$

Simply apply gradient ascent: $u \leftarrow u + \eta \nabla_u \mathcal{L}$

Rubric: 4 marks

Item	Marks
Correctly writes objective $\max_{\ u\ =1} \sum_i u^\top x_i $	1
Correctly forms Lagrangian with constraint $u^\top u = 1$	1
Correctly differentiates absolute value using $\text{sign}(u^\top x_i) x_i$ or subgradient	1.0
Correctly identifies gradient ascent rule to obtain u	1